# Complex Disordered Systems

Polymers and solvents

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# Today

- Good, poor and theta solvents
- Concentrated polymer solutions

# Polymers in solvents

- Polymers typically are dispersed in a medium (**solvent**).
- Solvent-polymer attractions (the **affinity**) influences the polymer conformations:
  - **good solvents** $\longrightarrow$ high affinity $\longrightarrow$ the polymer swells (expands)
  - **poor solvents** $\longrightarrow$ low affinity $\longrightarrow$ the polymer shrinks

  The affinity depends on the details of the molecular interactions but can be *coarse-grained*.

  We can construct a simple model with **effective interaction energies**
- $\varepsilon_{ss}$ for the solvent-solvent
- $\varepsilon_{\mathrm{pp}}$ for monomer-monomer
- $\varepsilon_{sp}$ for solvent-monomer interactions.

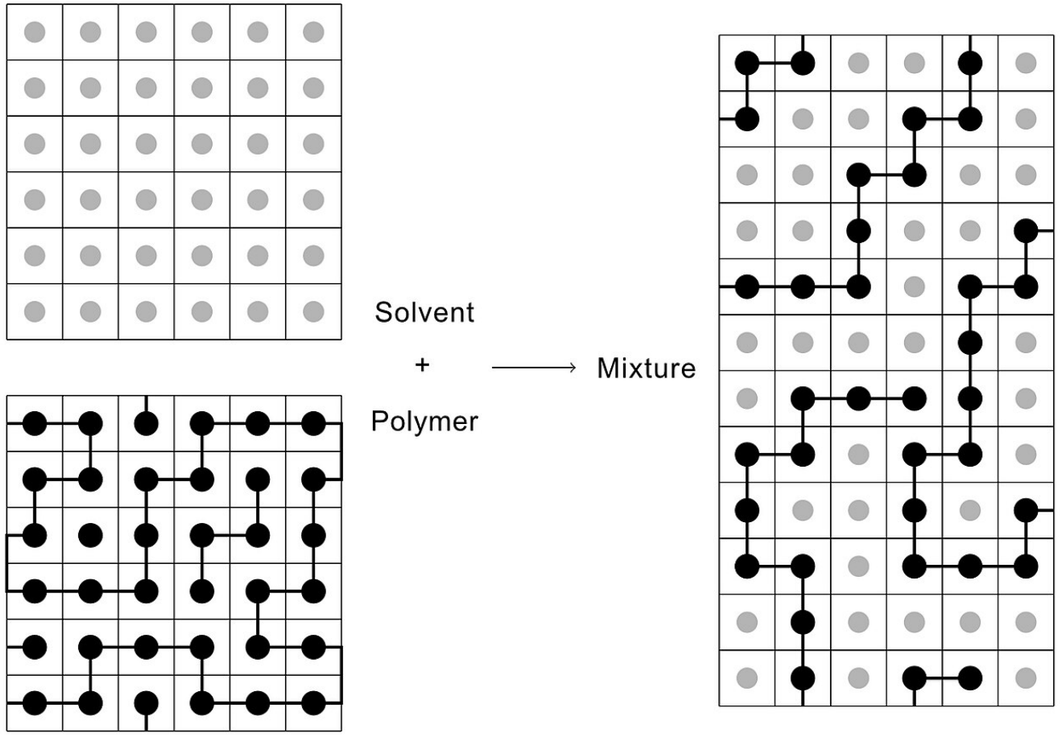
# Energy of mixing

**On-lattice model**:

- each site has $z$ nearest neighbours
- $N_s$ solvent molecules
- $N_p$ monomers
- $\mathrm{N_{sp}}$ solvent-monomer contacts.

**Effective interactions:**

- $\varepsilon_{ss}$ for the solvent-solvent
- $\varepsilon_{\mathrm{pp}}$ for monomer-monomer
- $\varepsilon_{sp}$ for solvent-monomer interactions.




Then the energy of mixing $\Delta U_{\mathrm{mix}}$ is obtained as the difference between the total energy $U$ of the mixed system and the sum of the energies of the **pure solvent** and **pure polymer**, $U_S + U_p$.

$$\Delta U_{\mathrm{mix}} = U - (U_S + U_p)$$

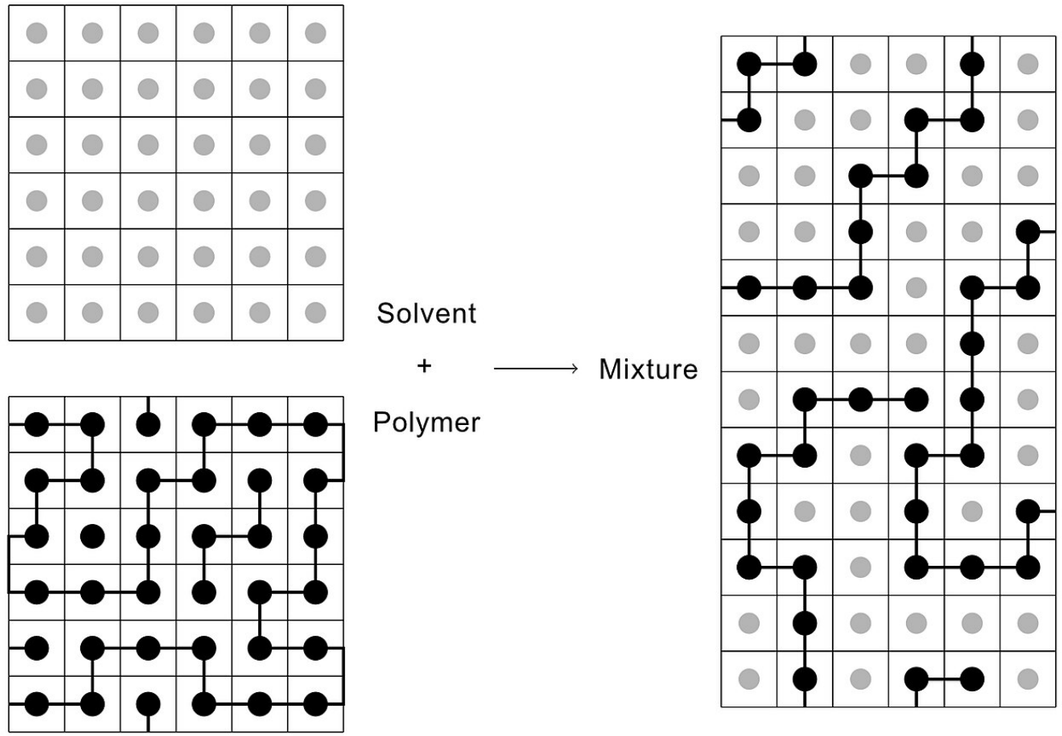
# Energy of mixing

- Energy of pure solvent

$$U_s = \frac{zN_s\varepsilon_{ss}}{2}$$

- Energy of pure polymer

$$U_p = \frac{zN_p\varepsilon_{pp}}{2}$$




**Energy of solution** $U = N_{sp}\varepsilon_{sp} + \frac{(zN_s - N_{sp})\,\varepsilon_{ss}}{2} + \frac{(zN_p - N_{sp})\,\varepsilon_{pp}}{2}$

We obtain that the **energy of mixing** $\Delta U_{\text{mix}} = N_{sp}\left[\varepsilon_{sp} - \frac{1}{2}\left(\varepsilon_{ss} + \varepsilon_{pp}\right)\right] \equiv N_{sp}\,\Delta\varepsilon$

which can **change sign** depending on our choices for the effective interaction energies.

# Excluded volume vs mixing energy

- The energy is not the whole story: each lattice site holds **one** molecule. This is the **excluded volume**, an **entropic** cost $\sim k_B T$ per monomer-monomer contact.
- The mixing energy works against it: for $\Delta\varepsilon > 0$ ($\Delta U_{\text{mix}} = N_{sp}\Delta\varepsilon > 0$), monomer-monomer contacts are favoured.
- The $\varepsilon$ of the Flory argument is an **effective contact free energy**

$$\varepsilon_{\text{eff}} \sim k_B T - z\,\Delta\varepsilon = k_B(T - \Theta), \qquad \Theta \sim \frac{z\,\Delta\varepsilon}{k_B}$$

# Good solvents, poor solvents and theta solvents

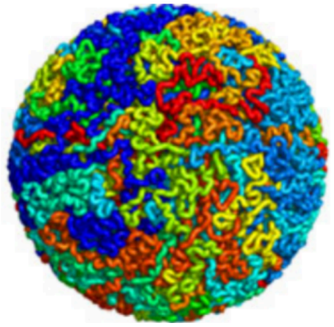
- **Good solvent**: $\varepsilon_{\text{eff}} > 0$ ($T > \Theta$)

Excluded volume wins and expands the chain. This includes $\Delta U_{\text{mix}} < 0$, but also weakly positive $\Delta U_{\text{mix}}$.

- **Poor solvent**: $\varepsilon_{\text{eff}} < 0$ ($T < \Theta$)

Requires $\Delta U_{\text{mix}} > 0$, with an attraction strong enough to overcome the excluded volume: the polymer collapses.

Collapsed chains: hydrophobic polypeptides, polystyrene in water, chromatin in the cell nucleus.



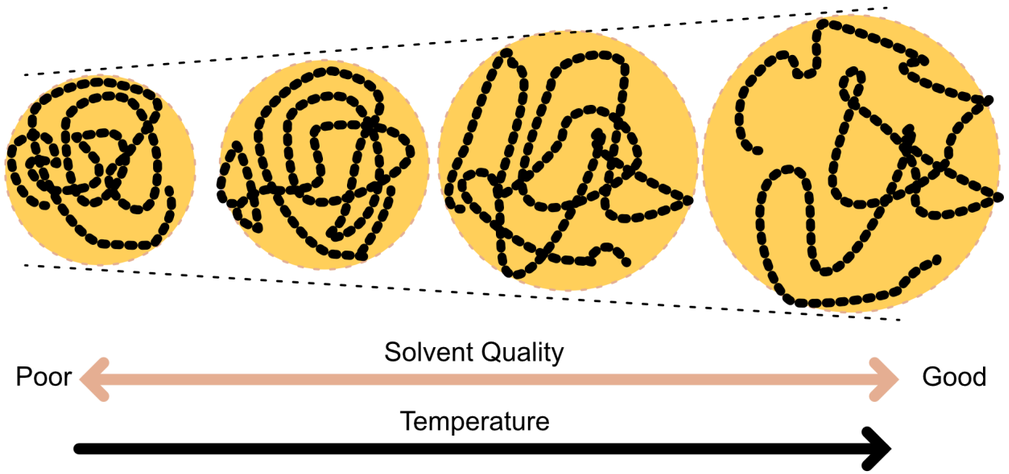
Chromatin globule model

- Tuning solvent properties (e.g. the temperature) leads to **conformational changes**

# Temperature dependence and theta ($\Theta$) solvents

The various affinities have temperature dependencies. Typically:

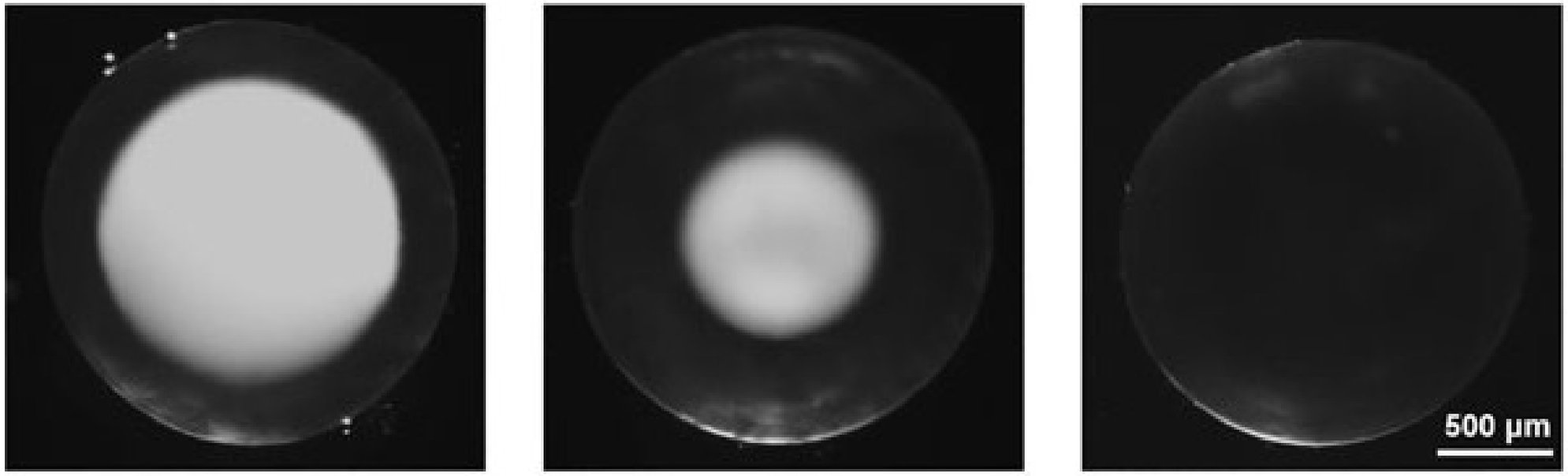
- At high $T$, the coil expand and the solvent is *good*
- At low $T$ the coil collapses and phase separation is observed (polymer-rich from polymer-poor phase)
- There is an intermediate temperature where the excluded volume and attractive interaction compensate each other ($\varepsilon_{\text{eff}} = 0$); in $d = 3$ the polymer then behaves like an ideal chain (freely jointed chain).
  - The temperature is conventionally named $\Theta$ **temperature** (or *Flory* temperature, after physicist Paul Flory)
  - The $\Theta$ temperature increases with the effective monomer-monomer attractions (we need higher temperature to reach ideal behaviour)




Swelling behaviour of polymers in solvent.

# Examples

- Example: **cold cataract**. The eye lens is a concentrated solution of proteins (crystallins) in water.
  - At body temperature water is a good solvent: the lens is transparent.
  - On cooling, water becomes a poorer solvent: the mixture separates into protein-rich and protein-poor regions, which scatter light, and the lens turns opaque.
  - Warming back to $37\,^{\circ}\mathrm{C}$ clears the lens within minutes: the change is reversible.




Mouse lenses at $4\,^{\circ}\mathrm{C}$, $25\,^{\circ}\mathrm{C}$ and $37\,^{\circ}\mathrm{C}$ (left to right). Images from Li *et al.*, *Front. Chem.* **9**, 767696 (2021), CC BY. Phase separation in cold cataract: Tanaka, Ishimoto & Chylack, *Science* **197**, 1010 (1977).

# Concentrated polymer solutions

- We mostly dealt with **single polymers** and their properties up to now.
- Many polymers together make a **polymer solution**

As we vary the **concentration** the behavior qualitatively changes.

We will increase the concentration in steps to reach **bulk polymers**

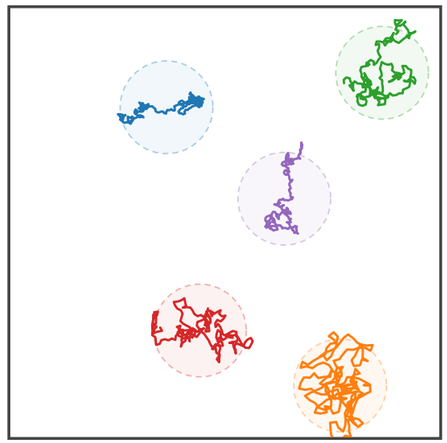
# Dilute regime

- The polymer coils are well-separated on average.
- The system is dilute if the polymer concentration $c$ is such that

$$\left(\frac{c}{M}N_A\right)\left(\frac{4\pi}{3}R_g^3\right) \ll 1$$

where $N_A$ is Avogadro's number, $M$ is the molar mass of a single chain and $R_g$ is the radius of gyration of the chain.

Essentially, the polymers **do not overlap**.



dilute c ≪ c*

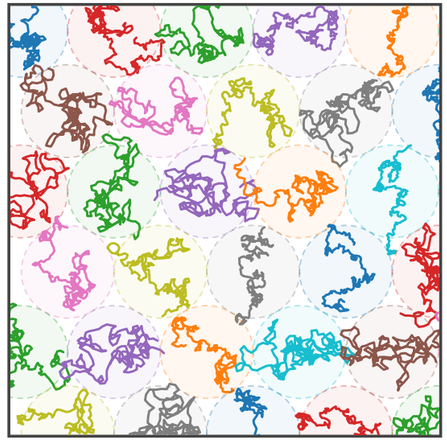
# Overlap concentration $c*$

Overlap occurs when the volume fraction of coils reaches unity and thus

$$\frac{c^*}{M} N_A \frac{4\pi}{3} R_g^3 \sim 1 \quad \therefore c^* = \frac{3M}{4\pi N_A R_g^3}$$

Phenomenologically one has $Rg = \langle R_g^2 \rangle^{1/2} = BM^\nu$ with an empirical exponent called the **Flory scaling exponent**. Hence

$$c^* = \frac{3}{4\pi N_A B^3} M^{1-3v}$$



overlap c ≈ c*

# Overlap concentration $c*$

$$c^* = \frac{3}{4\pi N_A B^3} M^{1-3v}$$

Example of polystyrene in a good solvent:

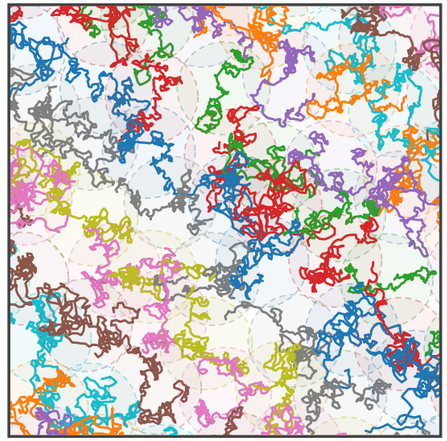
| Parameter | Value | Unit |
|---|---|---|
| Molar mass (M) | $10^6$ | g mol$^{-1}$ |
| Flory exponent (v) | 0.6 | |
| Prefactor (B) | 0.028 | nm (g mol$^{-1}$)$^{-0.6}$ |
| Overlap concentration (c*) | 0.29 | kg m$^{-3}$ (or 0.29 mg/ml) |
| Polystyrene density (ρ) | 1050 | kg m$^{-3}$ |
| Volume fraction of monomers | $0.28 \times 10^{-3}$ | |

For large polymers, $c^*$ can be very small.

# Semi-dilute

- The concentration is larger than the overlap concentration $c^*$, but still much smaller than the bulk density.
- The coils interpenetrate and **entanglement** begins
- Change in the dynamics (slowing down)
- Yet, the solution is still mostly solvent.

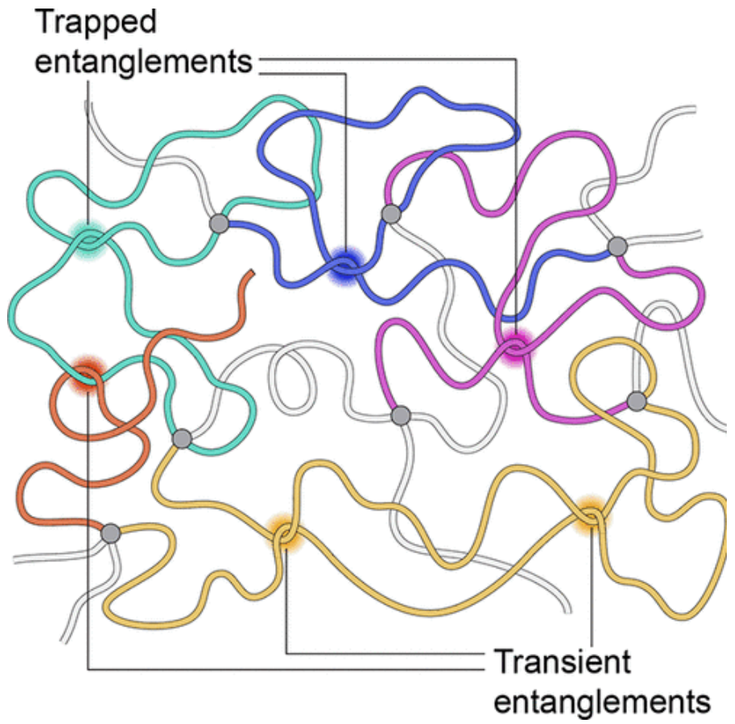
Start of the formation of a **polymer network** with transient **crowded regions** *within* the solvent.



semi-dilute c > c*

# Polymer entanglement

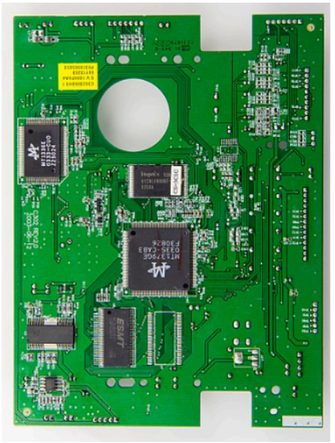
- Overlapping polymer chains become topologically **intertwined**
- Creates **physical knots** that restrict chain motion
  - changes flow behaviour (*rheology*)
  - changes material properties (e.g. elasticity, viscosity )
- Introduces **topological constraints** (transient/permanent)



# Concentrated: bulk polymers

- Polymer concentration approaches **bulk density**: **highly interpenetrated coils**, entanglements restrict segment motion, **solid-like** behaviour.
- **Cross-linking** forms **covalent bonds between separate polymer chains**, creating a **three-dimensional network**. Three main classes of bulk polymers, characterised by their degree of cross-linking:
  - **Elastomers** (rubbers): **low cross-linking** → flexible and elastic (natural rubber, silicone)
  - **Thermosets**: **high cross-linking** → rigid, hard, brittle network (epoxy resins)
  - **Thermoplastics**: **no cross-linking** → most everyday plastics (PE, PET, PS); behaviour upon cooling shows similarities to colloids



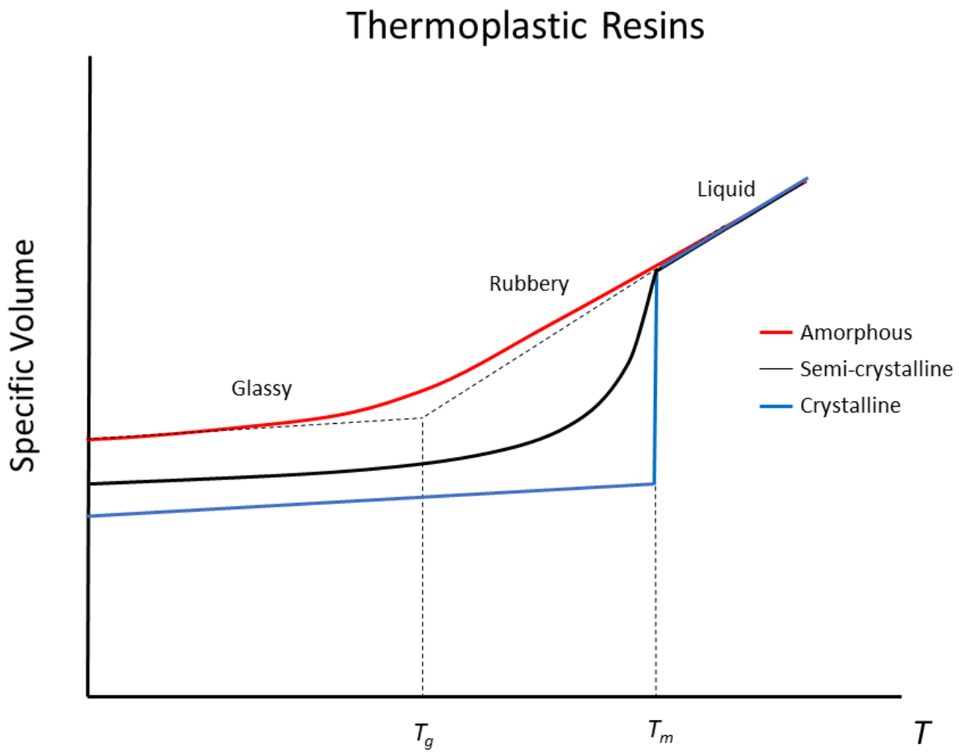




Natural rubber, epoxy substrate for circuits, PET used in plastic bottles

# Thermoplastics behaviour upon cooling

- At high temperature the free energy is dominated by **entropy**.
  - random assembly of intertwined flexible coils
- Upon cooling, potential energy dominates and chain rotation becomes restricted, favoring more extended configurations.
- Below melting temperature $T_m$, **crystals form** (lowest free energy state), but require slow cooling and significant molecular ordering.
- Rapid cooling below glass transition temperature $T_g$ ($< T_m$) **prevents crystallization**, forming an **amorphous glassy state** instead—a metastable, long-lived solid.
- Solid thermoplastics typically contain a **mixture of crystalline and amorphous regions**.



# Summary: polymers

| Topic | Key result |
|---|---|
| Freely-jointed chain | $\langle R^2 \rangle = N b_0^2$, Gaussian $P(\mathbf{R})$: a chain is a random walk |
| Radius of gyration | $\langle R_g^2 \rangle = \langle R^2 \rangle / 6$ |
| Freely-rotating chain | $\langle R^2 \rangle = C N b_0^2$, $C = \frac{1+\cos\theta}{1-\cos\theta}$: stiffness only changes the prefactor |
| Kuhn length | $b = C b_0$, $N' = N/C$: chemistry is hidden in $b$ |
| Excluded volume | $R \sim N^{3/5}$ (self-avoiding walk) |
| Solvent quality | $\varepsilon_{\text{eff}} \sim k_B(T - \Theta)$: excluded volume vs mixing energy; in $d = 3$, good $\nu \approx 3/5$; $\Theta$ $\nu = 1/2$; poor: collapsed globule $\nu = 1/3$ |
| Overlap concentration | $c^* = \frac{3M}{4\pi N_A R_g^3} \propto M^{1-3\nu}$, very small for long chains |
| Semi-dilute | $c > c^*$: interpenetration, entanglement, slow flow |
| Bulk polymers | elastomers, thermosets, thermoplastics; on cooling crystal (slow) or glass (fast) |

Full summary with links in the notes